# AMITY INSTITUTE OF INFORMATION TECHNOLOGY

Course code:MAT1013

Maths-LAB-21&22

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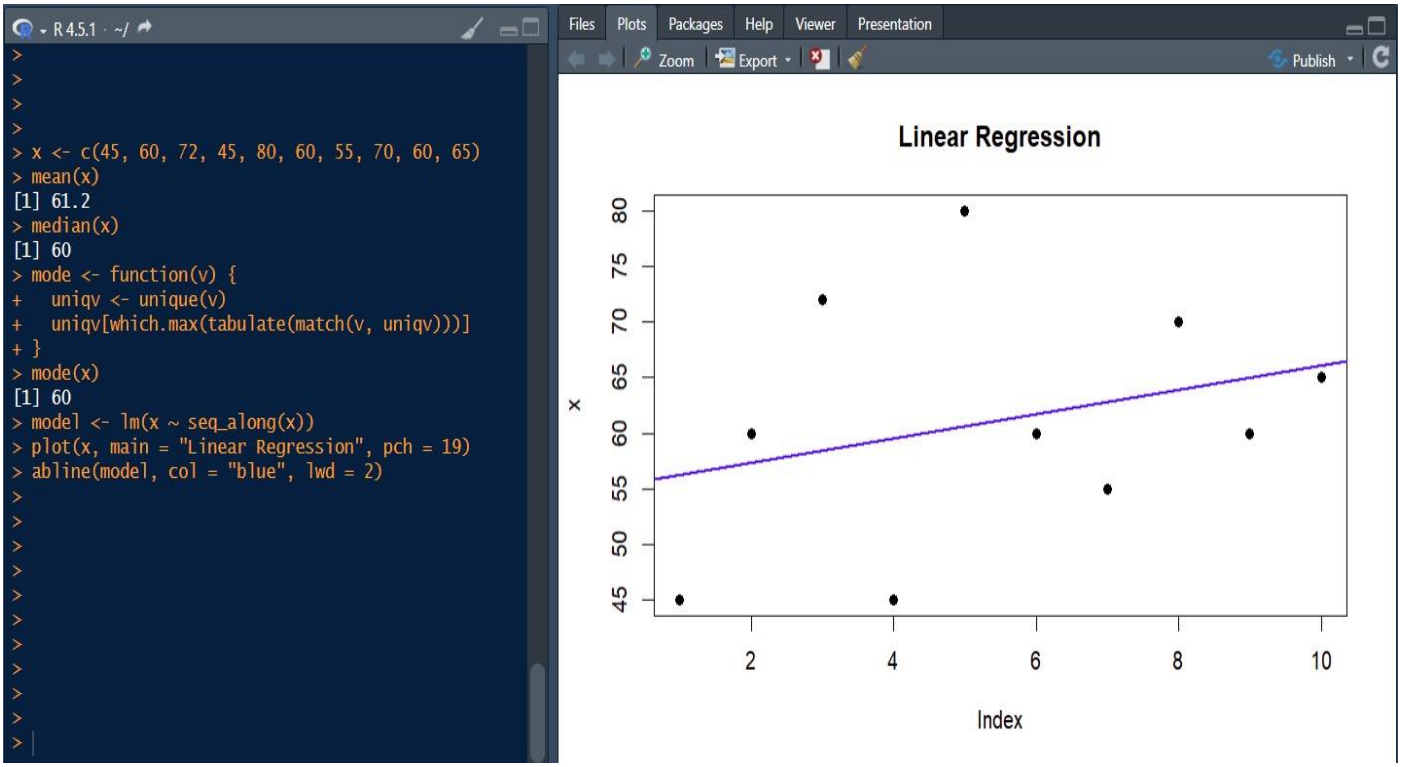
# 1)measure of central tendency(mean, median, mode)

```
x <- c(45, 60, 72, 45, 80, 60, 55, 70, 60, 65)
mean(x)
median(x)
mode <- function(v) {
  uniqv <- unique(v)
  uniqv[which.max(tabulate(match(v, uniqv)))]
}
mode(x)
model <- lm(x ~ seq_along(x))
plot(x, main = "Linear Regression", pch = 19)
abline(model, col = "blue", lwd = 2)
```

**<u>output:</u>**



## 2) measure of central dispersion(range, sd,cv)

wages <- c(1200, 1500, 1400, 1600, 1300, 1700, 1550, 1450)

range(wages)

diff(range(wages))

sd(wages)

cv <-(sd(wages) / mean(wages)) * 100

cv

**output:**

```
#madiha mufassira
#BCA
#A86904824018
wages <- c(1200, 1500, 1400, 1600, 1300, 1700, 1550, 1450)
range(wages)
diff(range(wages))
sd(wages)
cv <-(sd(wages) / mean(wages)) * 100
cv
```

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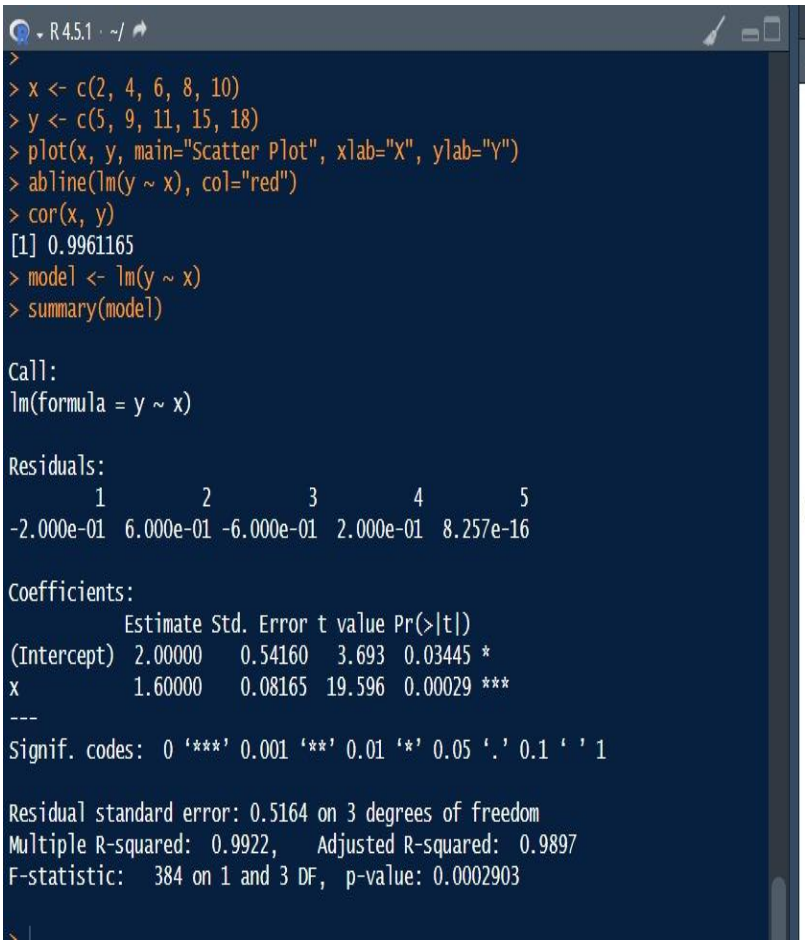
```
> wages <- c(1200, 1500, 1400, 1600, 1300, 1700, 1550, 1450)
> range(wages)
[1] 1200 1700
> diff(range(wages))
[1] 500
> sd(wages)
[1] 162.0185
> cv <-(sd(wages) / mean(wages)) * 100
> cv
[1] 11.07819
>
```

# 3)correlation, regression(using graph)

```
x <- c(2, 4, 6, 8, 10)

y <- c(5, 9, 11, 15, 18)

plot(x, y, main="Scatter Plot", xlab="X", ylab="Y")

abline(lm(y ~ x), col="red")

cor(x, y)

model <- lm(y ~ x)

summary(model)
```

## <u>output:</u>



```
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>
> x <- c(2, 4, 6, 8, 10)
> y <- c(5, 9, 11, 15, 18)
> plot(x, y, main="Scatter Plot", xlab="X", ylab="Y")
> abline(lm(y ~ x), col="red")
> cor(x, y)
[1] 0.9961165
> model <- lm(y ~ x)
> summary(model)

Call:
lm(formula = y ~ x)

Residuals:
        1         2         3         4         5
-2.000e-01  6.000e-01 -6.000e-01  2.000e-01  8.257e-16

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.00000    0.54160   3.693  0.03445 *
x            1.60000    0.08165  19.596  0.00029 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.5164 on 3 degrees of freedom
Multiple R-squared:  0.9922,	Adjusted R-squared:  0.9897
F-statistic:   384 on 1 and 3 DF,  p-value: 0.0002903
```

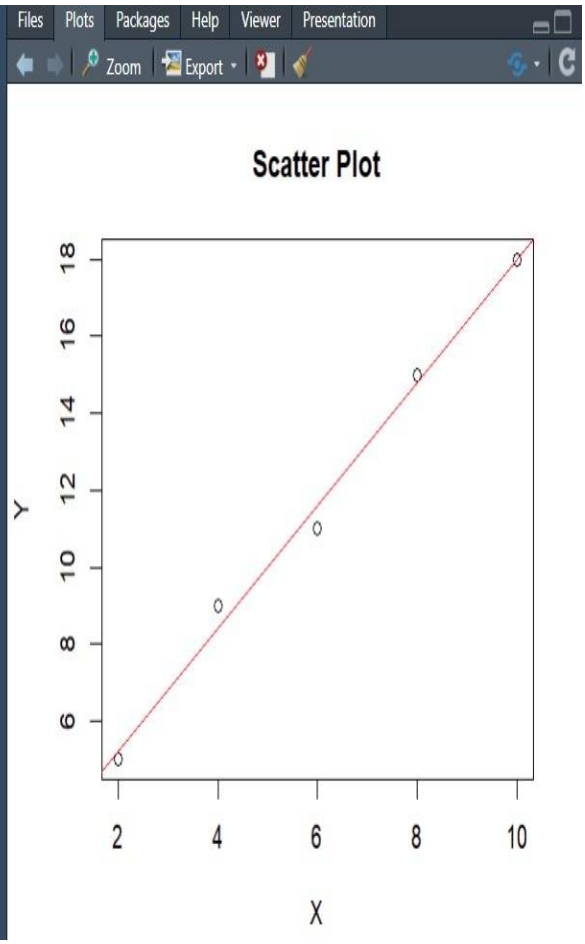

## 4)binomial and poisson distribution

dbinom(4, size = 6, prob = 0.5)

dpois(3, lambda = 2)

**<u>output:</u>**

```
#madiha mufassira
#BCA
#A86904824018
dbinom(4, size = 6, prob = 0.5)
dpois(3, lambda = 2)
```

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```
>
>
> dbinom(4, size = 6, prob = 0.5)
[1] 0.234375
> dpois(3, lambda = 2)
[1] 0.180447
>
```

## 5)t-test (single, two mean, paired)

x <- c(980, 1020, 1010, 1040, 1000, 990, 1030, 1015, 1025, 995)

t.test(x, mu = 1000)

class1 <- c(60, 62, 65, 68, 70)

class2 <- c(65, 67, 72, 74, 76)

t.test(class1, class2)

before <- c(70, 72, 68, 75, 71, 69)

after  <- c(74, 75, 70, 78, 73, 72)

t.test(before, after, paired = TRUE)

**<u>output:</u>**

**<u>single:</u>**

```
> x <- c(980, 1020, 1010, 1040, 1000, 990, 1030, 1015, 1025, 995)
> t.test(x, mu = 1000)

        One Sample t-test

data:  x
t = 1.7413, df = 9, p-value = 0.1156
alternative hypothesis: true mean is not equal to 1000
95 percent confidence interval:
  996.8592 1024.1408
sample estimates:
mean of x
   1010.5
```

## two mean:

```
> class1 <- c(60, 62, 65, 68, 70)
> class2 <- c(65, 67, 72, 74, 76)
> t.test(class1, class2)

        Welch Two Sample t-test

data:  class1 and class2
t = -2.0848, df = 7.8837, p-value = 0.0711
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -12.2320089   0.6320089
sample estimates:
mean of x mean of y
     65.0      70.8
```

## paired:

```
> before <- c(70, 72, 68, 75, 71, 69)
> after  <- c(74, 75, 70, 78, 73, 72)
> t.test(before, after, paired = TRUE)

        Paired t-test

data:  before and after
t = -9.2195, df = 5, p-value = 0.0002521
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -3.623320 -2.043347
sample estimates:
mean difference
      -2.833333
```

# 6)x^2-tesy (goodness of fit and independent attributes)

```
observed <- c(8, 10, 9, 11, 12, 10)

expected <- rep(10, 6)

chisq.test(x = observed, p = expected/sum(expected))

data <- matrix(c(30, 20, 25, 25),

          nrow = 2,

          byrow = TRUE)

rownames(data) <- c("Male", "Female")

colnames(data) <- c("Like", "Dislike")

chisq.test(data)
```

## <u>output:</u>

```
> observed <- c(8, 10, 9, 11, 12, 10)
> expected <- rep(10, 6)
> chisq.test(x = observed, p = expected/sum(expected))

        Chi-squared test for given probabilities

data:  observed
X-squared = 1, df = 5, p-value = 0.9626

> data <- matrix(c(30, 20, 25, 25),
+                nrow = 2,
+                byrow = TRUE)
> data <- matrix(c(30, 20, 25, 25),
+                nrow = 2,
+                byrow = TRUE)
> data <- matrix(c(30, 20, 25, 25),
+                nrow = 2,
+                byrow = TRUE)
> rownames(data) <- c("Male", "Female")
> colnames(data) <- c("Like", "Dislike")
> chisq.test(data)

        Pearson's Chi-squared test with Yates' continuity correction

data:  data
X-squared = 0.64646, df = 1, p-value = 0.4214
```